

# Treasury AI-Powered Alcohol Label Verification App

## Technical Planning Document v2.0

---

### 1. Recommended System Architecture

#### High-Level Architecture

Streamlit User Interface



CSV Upload Module



Image Upload Module



OCR Engine



Text Normalization



Record Matching Engine



Validation Engine



Recommendation Engine



Results Dashboard



CSV Export

---

## 2. Recommended Technology Stack

### User Interface

#### Streamlit

Reason:

- Fast prototype development
  - Built-in file upload controls
  - Built-in tables and dashboards
  - Simple deployment
  - Reduced frontend complexity
  - Well suited for proof-of-concept applications
- 

### Application Logic

#### Python

Reason:

- Strong OCR ecosystem
  - Excellent data-processing support
  - Rapid development
  - Large community support
-

# OCR

## Tesseract OCR

Reason:

- Runs locally
  - No cloud API dependency
  - Compatible with restricted network environments
  - Mature and widely adopted
  - Easily deployable
- 

# Text Matching

## RapidFuzz

Reason:

- Fast fuzzy matching
  - Local processing
  - Suitable for brand and class/type comparison
  - Lightweight and easy to maintain
- 

# Data Processing

## Pandas

Reason:

- CSV handling
  - Batch processing
  - Data validation
  - Result export
- 

# Image Handling

## **Pillow**

Reason:

- Lightweight image processing
  - Format validation
  - OCR preparation
- 

## **Deployment**

### **Render**

Reason:

- Simple deployment
- Free public URL
- Suitable for prototype evaluation
- Supports Python applications

Alternative:

Azure App Service

Potential future alignment with Treasury infrastructure.

---

# **3. Core Application Modules**

## **CSV Upload Module**

Purpose:

Accept application records used during validation.

Required Fields:

- record\_id
- brand\_name
- class\_type
- alcohol\_content

- net\_contents
- government\_warning

Optional Fields:

- bottler\_producer
- country\_of\_origin
- beverage\_type

Responsibilities:

- Validate CSV structure
  - Verify required fields
  - Display validation errors
  - Load records into memory
- 

## Image Upload Module

Purpose:

Accept one or more alcohol label images.

Supported Types:

- PNG
- JPG
- JPEG

Features:

- Multiple file upload
  - File validation
  - Image preview
  - Batch processing support
- 

## OCR Module

Purpose:

Extract readable text from uploaded label images.

Input:

Image file

Output:

Raw OCR text

Responsibilities:

- Process image
  - Run OCR
  - Return extracted text
- 

## **Text Normalization Module**

Purpose:

Standardize OCR output before matching.

Normalization may include:

- Case normalization
- Whitespace cleanup
- Common OCR correction
- Character cleanup

This improves matching reliability.

---

## **Record Matching Engine**

Purpose:

Identify the most likely application record for each uploaded label.

Inputs:

- OCR text
- Application records

Matching Factors:

- Brand Name
  - Class / Type
  - Alcohol Content
  - Net Contents
- 

## **Confidence Thresholds**

### **Auto Match**

95% and above

Action:

Automatically associate record.

---

### **Needs Review**

70%–94%

Action:

Display suggested match.

Human confirmation recommended.

---

### **No Match**

Below 70%

Action:

Flag for manual review.

---

## **Field Extraction Module**

Purpose:

Identify probable compliance values from OCR output.

Core Fields:

- Brand Name
- Class / Type
- Alcohol Content
- Net Contents
- Government Warning

Optional Fields:

- Bottler / Producer
  - Country of Origin
  - Beverage Type
- 

## Validation Engine

Purpose:

Compare expected values against detected values.

Validation Rules:

### **Brand Name**

Fuzzy Match

### **Class / Type**

Fuzzy Match

### **Alcohol Content**

Exact Match

### **Net Contents**

Exact Match

### **Government Warning**

Strict Match

---

# Recommendation Engine

Purpose:

Generate user-facing recommendations.

Possible Results:

## **Pass**

Detected value matches expected value.

## **Fail**

Detected value clearly conflicts with expected value.

## **Needs Review**

System confidence is insufficient for a reliable determination.

Examples:

- Poor OCR quality
  - Ambiguous matches
  - Missing information
- 

# Results Dashboard

Purpose:

Display verification results.

Displays:

- Image Name
  - Matched Record
  - Match Confidence
  - Validation Status
  - Processing Time
  - Notes
-

## CSV Export Module

Purpose:

Export verification results.

Output:

CSV file

Use Cases:

- Review workflows
  - Supervisor review
  - Audit support
- 

## 4. Validation Logic

### Exact Matching

Used when values must be legally or numerically precise.

Fields:

- Alcohol Content
- Net Contents

Example:

Expected:

45% Alc./Vol. (90 Proof)

Detected:

40% Alc./Vol. (80 Proof)

Result:

Fail

---

## Fuzzy Matching

Used when formatting differences should not cause failure.

Fields:

- Brand Name
- Class / Type

Example:

Expected:

Stone's Throw

Detected:

STONE'S THROW

Result:

Pass

---

## Strict Matching

Used when regulatory language must remain exact.

Fields:

- Government Warning

Result:

Fail if required warning language is missing or significantly altered.

---

## Normalized Matching

Used when formatting differs but meaning remains identical.

Example:

Expected:

750 mL

Detected:

750ml

Result:

Pass

Normalization may remove:

- Extra spaces
- Case differences
- Limited punctuation differences

Normalization does not ignore actual value differences.

---

## 5. Status Categories

### Pass

Detected field matches expected value.

---

### Fail

Detected field conflicts with expected value.

---

### Missing

Required field was not found.

---

### Needs Review

Insufficient confidence exists for an automatic determination.

Examples:

- OCR uncertainty
  - Ambiguous matching
  - Poor image quality
  - Missing information
- 

## Error

Processing failure occurred.

Examples:

- Invalid file
  - Corrupt image
  - Unexpected system exception
- 

# 6. User Interface Design

## Main Workflow

Step 1

Upload Application CSV

↓

Step 2

Upload Label Images

↓

Step 3

Run Verification

↓

Step 4

Review Results Dashboard



Step 5

Export Results

---

## Secondary Workflow

Single Label Test Mode

Used for:

- Testing
  - Demonstrations
  - Troubleshooting
- 

## 7. Results Dashboard Columns

- Image Name
  - Matched Record
  - Match Confidence
  - Field
  - Expected Value
  - Detected Value
  - Status
  - Confidence
  - Notes
  - Processing Time
- 

## 8. Batch Processing Design

Batch processing is a core workflow.

Prototype Target:

- 5–20 labels

Processing Method:

Sequential processing

Benefits:

- Simpler implementation
- Progressive results
- Better user feedback
- Easier troubleshooting

Results appear as each label completes processing.

Future versions may support:

- Queue processing
  - Parallel processing
  - 200–500+ label batches
- 

## 9. Error Handling

The application should handle:

- No CSV uploaded
- Missing required CSV fields
- No image uploaded
- Unsupported image type
- OCR failure
- Empty OCR result
- Record match failure
- Processing timeout
- Poor image quality

Error messages should be written in plain English.

Example:

"The image was uploaded, but no readable text was detected. Please try a clearer label image."

Example:

"The uploaded CSV is missing required fields: brand\_name, alcohol\_content."

---

## 10. Assumptions

This prototype:

- Does not connect to COLAs Online
  - Does not permanently store submitted labels
  - Does not permanently store application data
  - Does not make final legal compliance decisions
  - Assists human reviewers through decision support
  - Uses local OCR whenever possible
  - Minimizes dependency on external AI services
- 

## 11. Trade-Offs

### Chosen

Local OCR and rule-based validation.

Reason:

Supports restricted government environments where cloud access may be unavailable.

Provides predictable and explainable results.

---

### Not Chosen

Cloud-only AI vision APIs.

Reason:

- Network dependency

- Latency concerns
  - Cost concerns
  - Security concerns
  - Reduced reliability in restricted environments
- 

## Chosen

Batch-first workflow.

Reason:

Directly addresses stakeholder concerns regarding large application volumes.

---

## Not Chosen

Manual-entry-first workflow.

Reason:

Does not scale well and does not adequately address stakeholder pain points.

---

# 12. Definition of Done

The project is complete when:

- User can upload a CSV containing application records
- User can upload one or more label images
- OCR extracts readable text
- Record matching identifies likely application records
- App validates core required fields
- Results are displayed clearly
- Results can be exported as CSV
- Application is publicly deployed
- GitHub repository includes all source code
- README includes setup, run instructions, approach, assumptions, trade-offs, and limitations
- Sample labels and sample application datasets are included

- Human review remains the final authority for compliance decisions